85/100

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Duration: 60 Minutes** **Total Marks: 20**

Question Count: 4

Code of Conduct

I A.Vinay (192521441) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

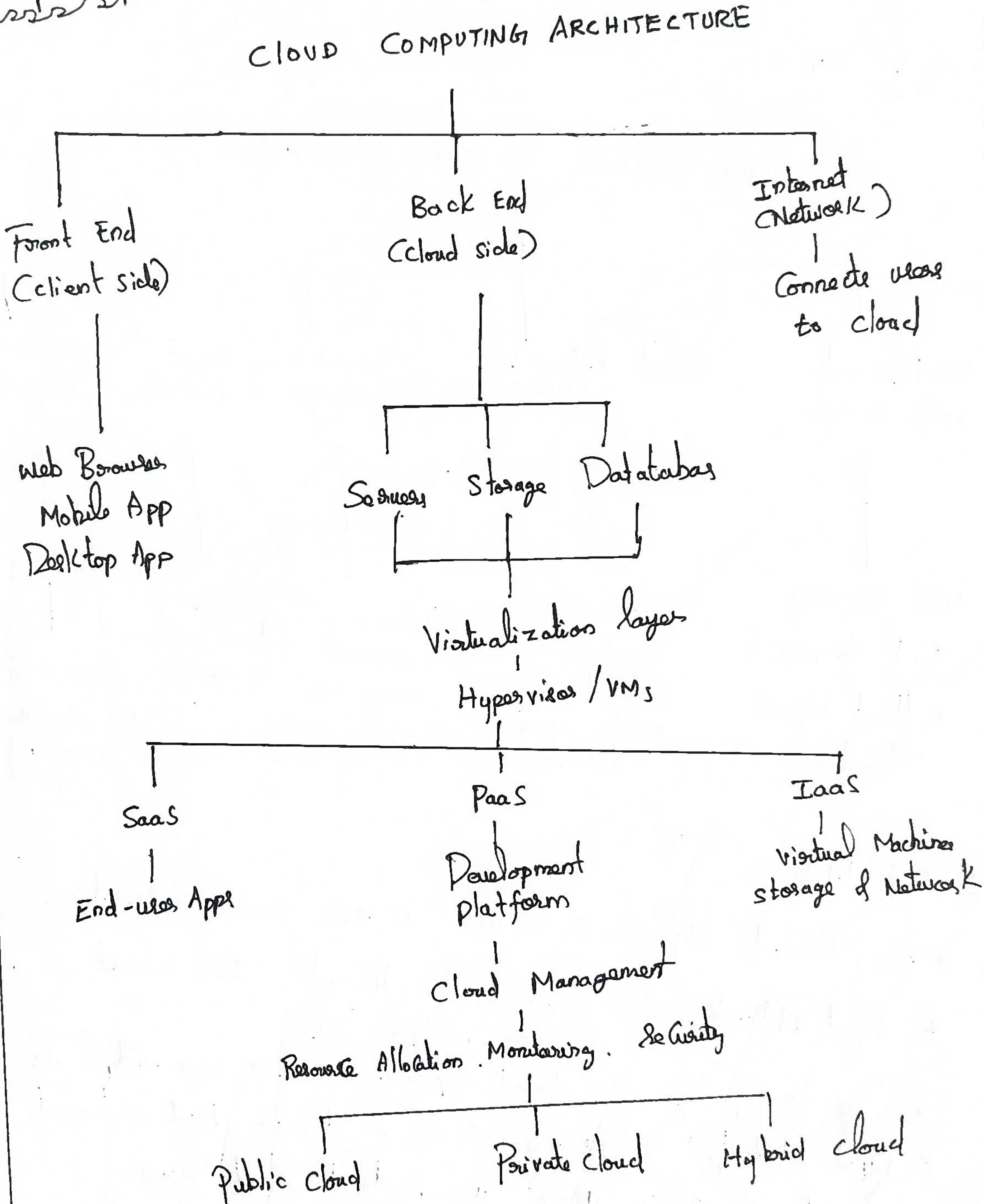
Signature of the student: Vinay

Criteria	Concept Identification (1.25)	Relationship Mapping (1.25)	Hierarchy and Organization (1)	Technical Relevance (1)	Visual Clarity (0.5)	TOTAL (5)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						

Signature of the Faculty **Total Marks Awarded:**

Cloud Computing Architecture

Concept map



2) Characteristics of cloud Computing

Concept map

CHARACTERISTICS OF CLOUD COMPUTING

- On-Demand self service
  - Users can use Cloud services without human interaction
- Broad Network Access
  - Access from anywhere via internet using mobile, laptop, desktop
- Resource Pooling
  - Shared resources among users
- Rapid Elasticity
  - Scale up or down quickly as needed
- Measure Servic
  - Pay only for resources used

Short labels

* On-Demand self service → users access services automatically through the internet
* Broad Network Access → Accessible from multiple devices
* Resource pooling → shared computing resources serve multiple users
* Rapid Elasticity → Resources increase or decrease based on demand
* Measured service → Usage is monitored and users Pay for what they use

Deployment Models ::

Concept Map ::

DEPLOYMENT MODELS

Public Cloud
- Owned by third-party Providers
- Usage Scenario
  - * website hosting
  - * Email services
  - * Startups
  - * App development

Private cloud
- Dedicated to one organization
- Usage Scenario
  - * Banking & finance
  - * Health care
  - * large enterprises

Hybrid cloud
- Combination of Public + private clouds
- Usage Scenario
  - * Business keeping sensitive data private
  - * cloud bursting

Community cloud
- Shared by organizations with similar needs
- Usage Scenario
  - * Government dept.
  - * universities
  - * research institutions

Goal:

Choose the deployment model based on security, cost, scalability and organizational needs

4) Cloud Benefits vs challenges

CLOUD COMPUTING

- Benefits
  - Cost efficiency
    - * Pay only for usage
    - * No large hardware investment
  - scalability
    - * Scale up or down easily
    - * Meets demand
  - Flexibility
    - * Access services anywhere anytime
    - * Supports remote work
  - Reliability
    - * High uptime with backups
    - * Global access
- Challenges
  - Security
    - * Risk of cyber attacks
    - * Data privacy
  - Downtime
    - * internet or server failures
    - * Services outage
  - Vendor Lock-in
    - * Difficult to switch provider
    - * Dependency on one vendor

Key Idea :

Cloud Computing offers Cost Savings, scalability and flexibility, but organizations must manage risks such as security, downtime and vendor lock-in.